TripMate: Collaborative Travel Planning & Management App

Submitted by

MANOHARAN K (230701177)

MOKESH M (230701192)

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BONAFIDE CERTIFICATE

Certified that this Project titled **TripMate: Collaborative Travel Planning & Management App** is the bonafide work of **Manoharan K (230701177)** and **Mokesh M (230701192)** who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Deepa B

Computer Science & Engineering

Rajalakshmi Engineering College

Thandalam, 602105

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Group travel planning is often hampered by the use of separate tools for budgeting, task allocation, scheduling, and communication, which leads to fragmented information and missed details. TripMate consolidates these disparate activities into a single, collaborative workspace where each trip serves as the central container for all planning elements. By anchoring expenses, checklist items, events, members, and notifications to one trip object, the app eliminates the need for users to toggle between multiple applications and ensures that every piece of data remains context-aware.

The user journey begins with the creation of a trip, which instantly becomes a shared environment that other participants can join. Within this space, users can add preparation tasks, log spending that is automatically summed, schedule activities with precise dates and times, and receive real-time alerts whenever any participant makes a change. This continuous feedback loop keeps the entire group synchronized without manual refreshes, fostering seamless coordination throughout the planning process.

By unifying all travel-planning functions in a single, persistent workspace, TripMate reduces cognitive load, minimizes the risk of oversight, and improves overall efficiency for group trips. The design also establishes a solid foundation for future enhancements such as itinerary recommendation, and integration with external travel services, demonstrating the viability of a centralized, trip-centric approach to collaborative travel management.

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CHAPTER 1

INTRODUCTION

1.1 General

Planning a journey with friends or family is fundamentally a social activity, yet the practicalities of coordination quickly become tangled. Groups normally resort to a patchwork of spreadsheets for budgets, chat applications for decisions, personal calendars for activities, and isolated note-taking tools for packing lists. The result is fragmented information, duplicated effort and inevitable mis-communication—expenses go untracked, tasks are forgotten, and schedules clash. TripMate is conceived as a direct antidote to this chaos. It treats each journey as a self-contained workspace—called a *Trip*—that gathers every planning artefact under a single roof: expenses, checklist tasks, events, member rosters and real-time notifications. By anchoring all actions to the same logical container, the app provides every participant with a shared, continuously synchronised view, dramatically reducing cognitive load and fostering transparent collaboration.

1.2 Scope of the Work

The present effort is limited to delivering a native Android application that embodies the trip-centric collaborative model described above. The user interface is built with Kotlin and Jetpack Compose. Core functional areas that fall within scope are:

- creating and managing multiple trips
- logging individual expenses and automatically aggregating totals
- scheduling events with precise date- and time-pickers
- inviting and managing members within a trip
- broadcasting real-time notifications for any change made by a collaborator.

Data persistence, user authentication and instantaneous synchronisation are supplied by Supabase, which offers a PostgreSQL backend, an Auth service and Realtime subscriptions with row-level security.

1.3 Aim and Objectives of the Project

The overarching aim of TripMate is to provide a single, unified workspace that enables groups to plan, organise and manage all aspects of a trip within one mobile application, thereby eliminating the reliance on disparate tools. First, a coherent data model is established in which a Trip is the primary container for Expenses, Checklist items, Events, Members and Notifications, guaranteeing that every piece of information remains contextually bound and that relational integrity is preserved throughout the life-cycle of the app. Second, a responsive user interface is crafted that follows Material 3 design principles, presenting a consistent dark-theme experience that scales gracefully on both small phones and larger tablets. Third, secure sign-up and sign-in capabilities are implemented using Supabase Auth, while row-level security ensures that users can only view or modify trips to which they have been invited. Fourth, the core functional modules—trip creation, expense entry with automatic total calculation, checklist handling, event scheduling, member invitation and notification delivery—are tightly integrated and made observable through Supabase's Realtime service. This guarantees that any modification performed by one participant propagates instantly to all other collaborators without manual refresh. Fifth, navigation across the eleven composable screens (splash, authentication, home, trip detail, expense, checklist, event, collaboration, notification, profile) is orchestrated with Navigation Compose, preserving back-stack integrity and providing smooth, intuitive transitions, including a complete stack clear-out on logout. Finally, the solution is validated through functional testing, usability evaluation and performance measurement on a representative set of Android devices.

CHAPTER 2

LITERATURE SURVEY

The growth of mobile technologies has significantly improved travel planning systems. Modern applications focus on integrating multiple functionalities such as location-based services, collaboration, and real-time data management to enhance user experience.

Mobile-based travel planning systems enable users to organize trips efficiently by providing features such as itinerary management and real-time updates [1]. These systems improve accessibility and convenience for travelers.

Collaboration is an important aspect of group travel. Research shows that collaborative mobile applications allow multiple users to work together, share updates, and coordinate tasks effectively, reducing communication gaps [2], [6].

Smart travel applications aim to combine multiple features into a single platform. These systems integrate trip planning, scheduling, and expense tracking, simplifying the overall process for users [3]. Such integration reduces dependency on multiple applications.

Location-based recommendation systems play a vital role in tourism applications. By using geographical data and user preferences, these systems suggest nearby tourist attractions and enhance travel experiences [4], [10].

Expense management systems are also essential for travel planning. Mobile applications allow users to track and manage expenses, ensuring better financial control and transparency, especially in group trips [5].

Cloud-based backend services provide scalable storage and real-time data synchronization. These systems enable efficient communication between the application and database, ensuring up-to-date information [9].

User experience design is another critical factor in mobile applications. A well-designed interface improves usability and user engagement by providing a smooth and intuitive workflow [8]. Additionally, modern programming frameworks such as Kotlin enhance performance and maintainability in mobile applications [7].

In summary, existing research emphasizes the integration of collaboration, location-based services, and data management in travel applications. However, most systems focus on individual features rather than offering a complete solution.

The proposed TripMate application addresses these limitations by combining all essential travel planning features into a unified system. It provides a collaborative environment where users can manage trips, track expenses, organize tasks, schedule events, and explore nearby locations within a single application. This integrated approach improves efficiency, enhances user experience, and offers a comprehensive solution for modern travel planning.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

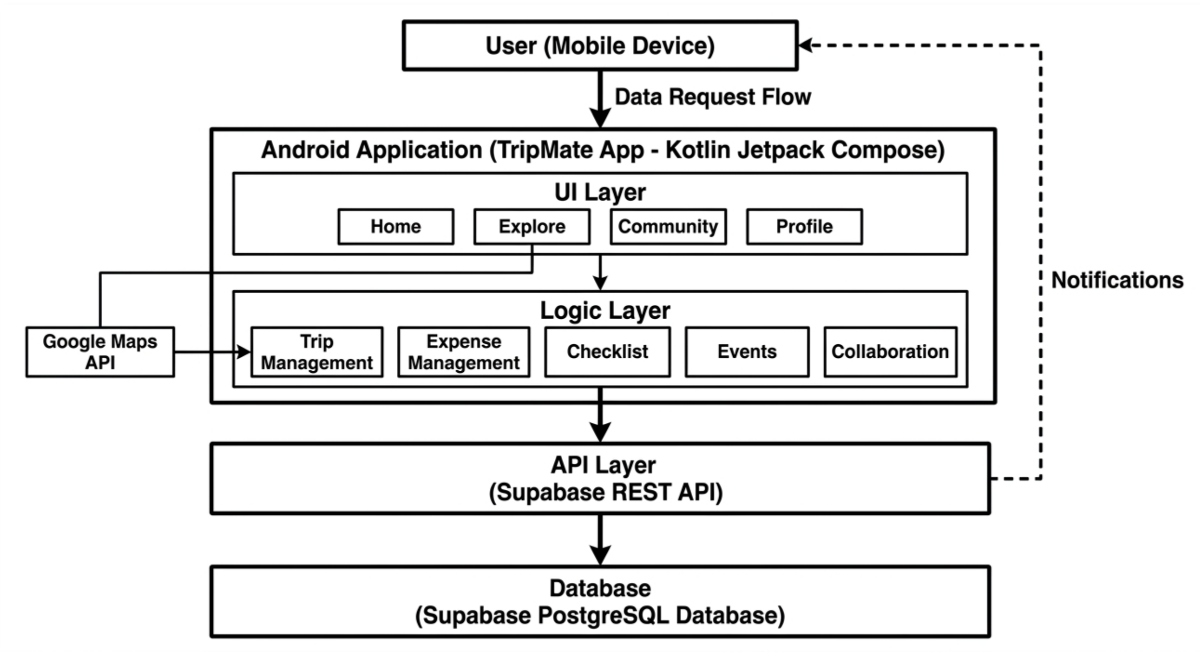
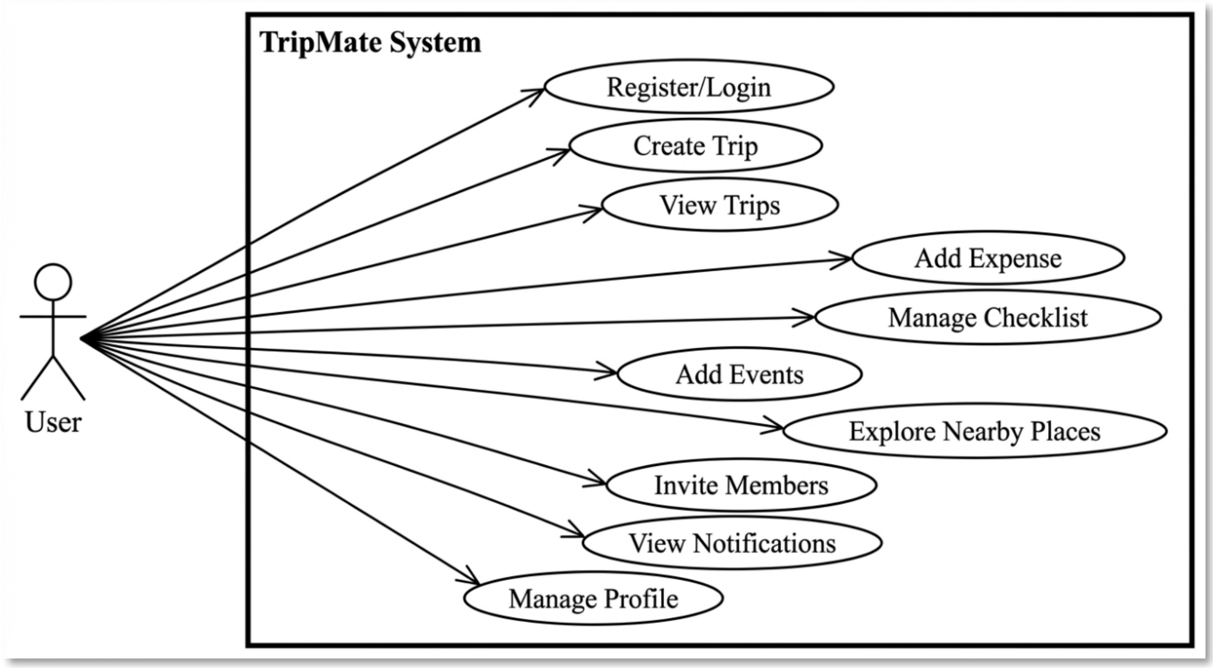


Fig 3.1.1 Architecture Diagram

The architecture of the TripMate application follows a layered approach consisting of the User Interface Layer, Application Logic Layer, API Layer, and Database Layer. The system is designed as a mobile application developed using Kotlin and Jetpack Compose. The user interacts with the application through a mobile device. The UI layer consists of core modules such as Home, Explore, Community, and Profile, which handle user interaction and navigation. The application logic layer manages core functionalities including trip management, expense tracking, checklist handling, event scheduling, and collaboration features. This layer processes user input and coordinates data flow between the UI and backend. The API layer connects the application to the backend services using Supabase REST APIs. It enables data communication such as fetching trips, storing expenses, and managing user activities. The database layer is implemented using Supabase PostgreSQL, where all structured data such as users, trips, expenses, checklist items, events, and notifications are stored.

3.2 USE CASE DIAGRAM



The use case diagram represents the interactions between the user and the TripMate system. The primary actor in the system is the user, who interacts with various functionalities provided by the application. The user can register or log in to access the system. Once authenticated, the user can create and manage trips, view trip details, and track expenses. The checklist management feature allows users to add, update, and mark tasks as completed. Users can also add events to plan their itinerary effectively. The Explore feature enables users to discover nearby tourist places based on their location. The collaboration feature allows users to invite members to a trip and work together on planning. Notifications keep the user informed about activities such as new expenses, tasks, or member additions. Additionally, the user can manage their profile, view travel statistics, and update personal preferences. The use case diagram highlights the comprehensive functionality of the system and demonstrates how users interact with different modules of the application.

3.3 HARDWARE SPECIFICATIONS

The TripMate application is designed to operate on standard modern computing devices and mobile platforms.

Development System Requirements

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free space
- Operating System: Windows 10 / macOS / Linux
- Development Tools: Android Studio / VS Code

Mobile Device Requirements

- Android Version: Android 8.0 (Oreo) or above
- RAM: Minimum 4 GB
- Storage: Minimum 100 MB free space
- Internet Connectivity: Required for API communication
- GPS Sensor: Required for location-based features

Server Requirements

- Backend: Supabase (Cloud-based)
- Database: PostgreSQL
- API Access: Internet-based REST APIs

The hardware requirements ensure smooth performance of the application, efficient data handling, and seamless user experience across supported devices.

3.4 SOFTWARE SPECIFICATIONS

The TripMate application is developed using modern software technologies to ensure efficiency, scalability, and a smooth user experience. The software stack consists of frontend development tools, backend services, database management systems, and supporting APIs.

Frontend Software Requirements

- Programming Language: Kotlin
- UI Framework: Jetpack Compose
- Architecture: Single Activity Architecture with Navigation Compose
- Design System: Material Design 3 (Dark Theme)
- Development Environment: Android Studio / Visual Studio Code

The frontend is responsible for rendering the user interface and handling user interactions such as navigation, input handling, and displaying dynamic data.

Backend Software Requirements

- Backend Platform: Supabase
- API Type: RESTful APIs
- Authentication: Supabase Authentication (Email/Password-based)

The backend manages data storage, user authentication, and communication between the mobile application and the database.

Database Requirements

- Database System: PostgreSQL (provided by Supabase)
- Data Storage: Cloud-based relational database

The database stores structured data such as user profiles, trips, expenses, checklist items, events, notifications, and travel memories.

Third-Party APIs and Services

- Maps and Location Services: Google Maps API
- Places and Images: Google Places API

These APIs are used to fetch real-time location data, display maps, and provide nearby tourist place information.

Version Control

- Platform: Git
- Repository Hosting: GitHub

Version control is used to manage source code, track changes, and enable collaborative development.

Operating System Requirements

- Development: Windows / macOS / Linux
- Deployment: Android Operating System (Version 8.0 and above)

Other Supporting Tools

- Package Manager: Gradle
- API Testing Tools: Postman (optional)
- Emulator: Android Emulator / Physical Device

CHAPTER 4

PROPOSED SYSTEM

The proposed system, TripMate: Collaborative Travel Planning & Management App, is designed to simplify and enhance the process of planning and managing trips in a collaborative environment. The system addresses the limitations of traditional travel planning methods, where users rely on multiple disconnected applications for booking, expense tracking, task management, and communication. TripMate integrates all these functionalities into a single unified platform, providing a seamless and efficient user experience.

The core concept of the proposed system revolves around a centralized Trip entity, which acts as the primary workspace for all travel-related activities. Each trip contains interconnected modules such as expense management, checklist tracking, event scheduling, collaboration, and notifications. This hierarchical structure ensures that all data related to a specific trip is organized and easily accessible, reducing complexity and improving usability.

The system enables users to create trips, define destinations, set budgets, and plan itineraries in an intuitive manner. Through the checklist module, users can manage tasks such as packing, bookings, and documentation. The expense module allows users to record and monitor spending in real time, providing transparency and better financial control. The event scheduling feature helps users organize their itinerary by adding time-based activities and travel plans.

One of the key innovations of the proposed system is its collaborative functionality. Users can invite members to a trip and collectively manage tasks, expenses, and events. All activities performed by members are reflected in a shared environment, ensuring coordination and reducing miscommunication. The system also includes a notification mechanism that provides real-time updates whenever changes occur within a trip, such as adding expenses or completing tasks.

Additionally, the proposed system incorporates a location-based exploration feature that enables users to discover nearby tourist places. By utilizing device location and map services, users can identify popular destinations and integrate them directly into their travel plans. This feature enhances the overall travel experience by combining planning and discovery in a single platform.

The system is designed with a modern mobile architecture using Kotlin and Jetpack Compose, ensuring a responsive and interactive user interface. Backend services are managed using a cloud-based platform, allowing secure data storage and efficient data retrieval. The modular design ensures scalability, enabling the system to accommodate additional features in the future.

Overall, the proposed system offers a comprehensive solution for travel planning by combining multiple functionalities into a single application. It improves efficiency, enhances collaboration, and provides a structured approach to managing trips. By addressing the challenges of fragmented tools and lack of coordination, TripMate aims to deliver a smarter and more organized travel planning experience.

CHAPTER 5

MODULE DESCRIPTION

The TripMate application is structured into several functional modules, each responsible for handling specific aspects of the travel planning process. These modules are designed to work together in an integrated manner to provide a seamless user experience.

5.1 User Authentication Module

This module manages user registration and login functionalities. It allows users to securely create an account and access the application using their credentials. The module ensures that only authorized users can access the system and perform operations.

5.2 Trip Management Module

The Trip Management module is the core component of the system. It enables users to create, view, update, and manage trips. Users can specify trip details such as destination, duration, and budget. Each trip acts as a central workspace where all related activities like expenses, tasks, and events are organized. This module ensures proper structuring and management of travel data.

5.3 Expense Management Module

This module allows users to track and manage expenses related to a trip. Users can add expense details such as title, amount, and payer information. The system calculates the total expenditure and provides a clear overview of spending. This helps users maintain financial control and transparency throughout the trip.

5.4 Checklist Management Module

The Checklist module helps users manage tasks required for the trip. Users can add tasks, mark them as completed, and track progress. This module ensures that important activities such as packing, booking, and preparation are not overlooked.

5.5 Event Scheduling Module

The Event module is used to plan and organize the trip itinerary. Users can add events with details such as date, time, and location. This helps in structuring the daily schedule of the trip. The module allows users to visualize and manage their travel plan effectively.

5.6 Collaboration Module

The Collaboration module enables multiple users to work together on a single trip. Users can invite members to join a trip and participate in planning. All members can contribute by adding expenses, tasks, and events. This module ensures better coordination and shared responsibility among travelers.

5.7 Explore Module

The Explore module allows users to discover nearby tourist places using location-based services. It utilizes map and location APIs to fetch and display popular destinations. Users can view details of places and add them to their trip plan. This module enhances the travel experience by integrating discovery with planning.

5.8 Profile Module

The Profile module manages user-related information and settings. It displays user details, travel statistics, and preferences. Users can view their past trips, manage personal data, and log out of the application. This module provides a personalized experience and control over user data.

CHAPTER 6

OUTPUT

Fig.6.1: Login Screen

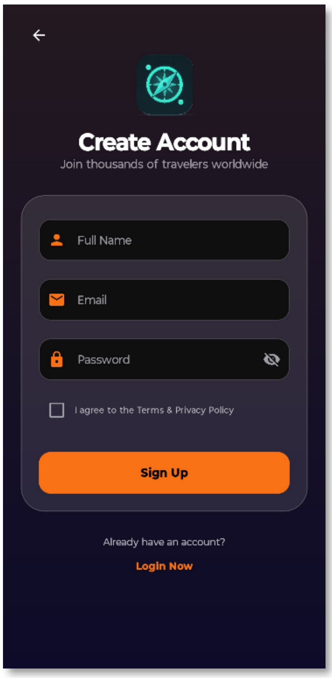


Fig.6.2: SignUp Screen

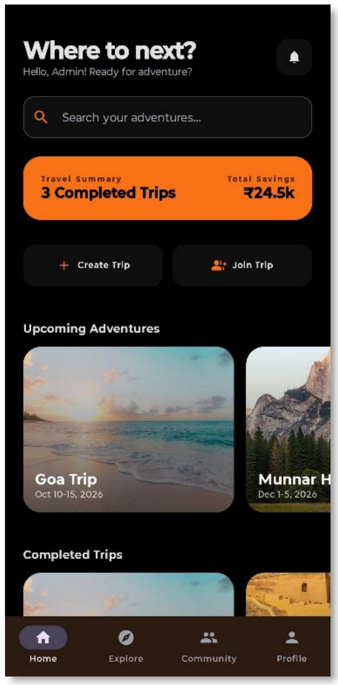


Fig.6.3: Home Screen

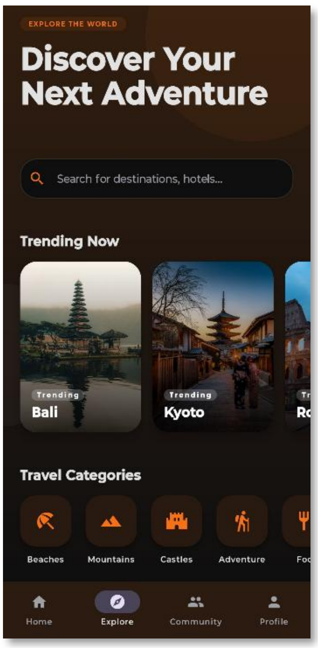


Fig.6.4: Explore Screen

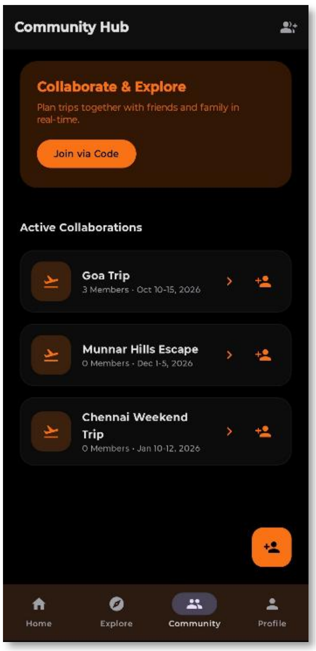


Fig.6.5: Collaborative Screen

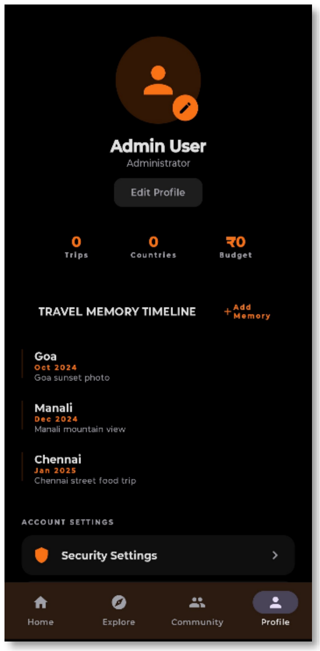


Fig.6.6: Profile Screen

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion:

The TripMate application has been successfully developed as a comprehensive solution for collaborative travel planning and management. The system effectively integrates multiple functionalities such as trip management, expense tracking, checklist organization, event scheduling, and member collaboration into a single unified platform. This integration eliminates the need for multiple separate tools and simplifies the overall travel planning process.

The application is designed with a user-friendly interface and structured workflow, allowing users to easily create trips, manage activities, and track progress. The centralized trip-based architecture ensures that all related information is organized efficiently, improving accessibility and usability. The collaboration feature enables multiple users to participate in planning, thereby enhancing coordination and reducing communication gaps.

The inclusion of real-time notifications ensures that users remain updated with all activities within a trip, promoting transparency and accountability among members. Additionally, the location-based exploration feature enhances user experience by allowing users to discover nearby tourist destinations and integrate them into their travel plans.

Overall, the TripMate application successfully addresses the challenges associated with traditional travel planning methods by providing a structured, efficient, and collaborative environment. The system demonstrates reliability, scalability, and practical usability, making it a valuable tool for modern travel planning.

Future Enhancements:

Although the current system provides a wide range of features, several improvements can be implemented in the future to enhance functionality and user experience.

One major enhancement is the integration of real-time multi-user synchronization using advanced backend services, allowing seamless updates across multiple devices simultaneously. This will further strengthen collaboration capabilities.

The system can also be extended by incorporating advanced analytics and insights, such as personalized travel recommendations, budget optimization suggestions, and predictive planning based on user behavior.

Integration with external services such as hotel booking, transportation systems, and weather forecasting APIs can provide a more comprehensive travel solution. This would allow users to plan their entire journey within a single application.

Another potential improvement is the addition of offline functionality, enabling users to access trip details and manage tasks even without an internet connection. Data synchronization can occur once connectivity is restored.

The application can also be enhanced by introducing multimedia support in the travel memory module, including video storage and sharing capabilities. Additionally, implementing social sharing features would allow users to share their travel experiences on various platforms.

Finally, improvements in security, such as advanced authentication mechanisms and role-based access control, can be incorporated to ensure better data protection and privacy.

CHAPTER 8

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